SECTION B

Answer **all** questions.

7. (a) The boiling temperatures of the hydrides of some Group 4 and Group 7 elements are shown in the table below.

Group 4 hydride	Boiling temperature / °C	Group 7 hydride	Boiling temperature / °C
CH_4	–161	HF	20
SiH_4	–112	HCl	–85
GeH_4	–88	HBr	–66

Hydrogen has an electronegativity value of 2.1.

The electronegativity values of the Group 4 and Group 7 elements are given below.

Element	Electronegativity	Element	Electronegativity
C	2.5	F	4.0
Si	1.8	Cl	3.0
Ge	1.8	Br	2.8

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For the hydrides of the Group 4 and Group 7 elements shown in the table opposite

- describe any trends and anomalies in boiling temperature
- explain any differences in boiling temperature in terms of the intermolecular forces present.

[6 QER]

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(b) Methane reacts with copper(I) oxide according to the following equation.

$$\ldots\ldots CH_4 + \ldots\ldots Cu_2O \longrightarrow \ldots\ldots Cu + \ldots\ldots CO_2 + \ldots\ldots H_2O$$

(i) Balance the equation. [1]

(ii) Explain why this reaction is described as a redox process. [1]

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(c) Compounds containing a Group 1 metal, a Group 3 metal and hydrogen only are known as complex metal hydrides. There has been much interest in their use as hydrogen storage systems for future fuel cell-powered vehicles.

One such hydride contains 38.7% Li and 50.1% Al.

Find the empirical formula of this compound. [2]

Empirical formula ....

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